

Monalisa Dokania

Software Engineer

Location: Tempe, Arizona, USA · Open to relocate

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PROFESSIONAL PROFILE

Software engineer focused on **distributed systems, cloud infrastructure, and applied AI**. I build production-ready backend platforms, event-driven systems, and thoughtfully designed LLM-powered applications.

Currently a founding engineer at an early-stage startup; alongside, I build and publish agentic AI systems.

TECHNICAL SKILLS

Languages	Python · Go · TypeScript · Java · JavaScript
Backend	FastAPI · Node.js · tRPC · Spring Boot · PostgreSQL · Redis · Kafka · Microservices
Frontend	React · Next.js · Vite · Tailwind · HTML/CSS
AI / ML	Agentic AI · RAG · LangChain / LangGraph · RAGAS · Prompt Engineering · Vector Databases
Infrastructure	AWS · EKS · Kubernetes · Docker · Terraform · Jenkins · Prometheus & Grafana
Tools	Git · GitHub Actions · JIRA · Linux · CI/CD · Load Testing

WORK EXPERIENCE

06/2024 – present

Scottsdale, AZ, USA

Lead Software Engineer

BLUSVN Technologies

- **Founding engineer** on a distributed water-delivery platform; owned the full product lifecycle from architecture to production across **backend, infrastructure, and ML**.
- Architected and shipped **100+ backend APIs** (TypeScript/tRPC, PostgreSQL) powering customer, driver, and admin applications; reduced end-to-end latency by **40%**.
- Engineered high-performance **Golang microservices** with concurrent worker pools and batched notification delivery for order allocation and driver dispatch workflows.
- Built **Kafka**-based location workflows to support responsive delivery status updates.
- Built real-time **ML inference pipelines** for face and ID validation, and **AI-assisted agent workflows for log analysis** that cut manual investigation effort during incident triage.
- Designed and operated production **AWS/EKS infrastructure** with Terraform; reduced AWS spend by **12%** and led **load and chaos testing** to validate SLOs under peak load.
- Partnered with product managers and stakeholders on technical architecture and **led engineering delivery**; managed **CI/CD** and Agile sprints.

05/2023 – 05/2024

Tempe, AZ, USA

Graduate Teaching Assistant

Arizona State University

- Supported Software Verification, Validation & Testing — **900+ students** across sections.
- Guided students in unit testing, debugging, code reviews, and OOP design principles.
- Led lab sessions and office hours; developed course materials and grading rubrics.

07/2016 – 05/2022

India

Business Operations Analyst

M/S Rajiv Industries - Family-owned manufacturing business

- Supported day-to-day operations and reporting workflows to track sales, inventory, customer orders, and SKU-level performance at a family-run manufacturing business.
- Consolidated historical sales and inventory records into structured internal datasets, to support data retrieval, recurring demand analysis and stock-planning visibility.
- Returned to full-time technical work through an M.S. in Computer Science in 2022.

06/2015 – 06/2016

Noida, India

Associate Software Engineer

Indus Valley Partners Pvt. Ltd.

- Developed **custom applications on the Geneva financial platform**, a portfolio management system used by asset managers.
- Contributed to a major platform upgrade (Geneva 8.0 → 15.2), including data migration validation and statistical reconciliation across portfolios.

TECHNICAL PROJECTS

04/2026 – present

Live Suggestions Copilot: Real-time AI Meeting Assistant

Built a real-time AI copilot to transcribe live meeting audio and generate context-aware suggestions, with a streaming chat panel for follow-up answers. Added error handling & engineered prompt rules for topic locking, recency weight, anti-repetition, suggestion diversity & hallucination control.

Tech Stack: Next.js, TypeScript, Groq, Whisper, MediaRecorder API, SSE, Tailwind

 <https://github.com/msdokania/live-suggestions-copilot>

 **Live demo:** <https://live-suggestions-copilot.vercel.app/>

12/2025 – present

Research Assistant — Hybrid RAG + RAGAS Evaluation

Production-grade research assistant as part of an Agentic AI Series: Implemented hybrid retrieval using semantic search + BM25, cross-encoder re-ranking, citation enforcement, and RAGAS-based evaluation against ground-truth Q&A pairs to measure retrieval quality.

 <https://github.com/msdokania/agentic-ai-series/tree/master/research-assistant-rag>

EDUCATION

08/2022 – 05/2024

Tempe, AZ, USA

Master of Science, Computer Science

Arizona State University

Final grade: 4.0 / 4.0

2011 – 2015

Allahabad, India

Bachelor of Technology, Information Technology

Motilal Nehru National Institute of Technology (MNNIT Allahabad)